

High-dimensional Two-sample Tests via Max-sliced Wasserstein Distance and Bootstrapping

Theory and Reproduction Study

Biostatistics Final Project

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Based on Hu and Lin, *Biometrika*, 2025.

Part I: Theory

- Two-sample distribution testing in high dimension
- Classical routes and Wasserstein-based alternatives
- Sparse max-sliced Wasserstein statistic
- Bootstrap calibration and simultaneous confidence intervals
- Computation: one-dimensional W_1 and ℓ_1 screening
- Theoretical validity: Gaussian bridge, size control, and power

Part II: Reproduction and Discussion

- Reproduction overview and experimental pipeline
- Model A simulation: sparse decaying mean shifts
- MSWD-only bootstrap diagnostic
- Real data: LGG vs GBM DNA methylation
- Global rejection, marginal genes, and GO-term localization
- QQ plots, limitations, future work, and references

Part I: Theory

From distributional comparison to sparse max-sliced Wasserstein testing

Two-sample Distribution Testing

Given two independent samples

$$X_1, \dots, X_{n_1} \stackrel{i.i.d.}{\sim} P, \quad Y_1, \dots, Y_{n_2} \stackrel{i.i.d.}{\sim} Q,$$

where P, Q are distributions on $\mathbb{R}^p$, the goal is to test

$$H_0 : P = Q \quad \text{versus} \quad H_1 : P \neq Q.$$

Interpretation

This is a global distributional test: rejecting H_0 means that the two populations differ in some aspect of their distributions, not necessarily only in their means.

- p may be comparable to, or much larger than, the sample size.
- The difference may appear in means, variances, tails, dependence structure, or higher-order moments.

Classical Routes for Distributional Comparison

Method family	Main idea	Limitation
Univariate KS / CvM / run tests	Compare empirical distribution functions or ranks	Mainly one-dimensional; no natural total ordering in $\mathbb{R}^p$
Density / density ratio	Estimate $p(x)$, $q(x)$, or $p(x)/q(x)$, often via kernel smoothing	Severe curse of dimensionality in density estimation
Graph / nearest neighbor	Check whether labels mix on a pooled-sample graph	Graph structure may become unstable in high dimension
Energy distance / BG	Compare within-group and between-group distances	Sparse signals may be diluted by irrelevant coordinates
MMD	Compare kernel mean embeddings in an RKHS	Sensitive to kernel and bandwidth; weak for sparse or high-order differences

Common difficulty: The discrepancy can be weak, nonlinear, high-order, or sparse; when only a few coordinates are informative, full-dimensional methods may dilute the signal by mixing them with many irrelevant coordinates.

A Wasserstein Route for Distributional Comparison

The Wasserstein distance is defined as

$$W_q(P, Q) = \left(\inf_{\gamma \in \Gamma(P, Q)} \int \|x - y\|_2^q d\gamma(x, y) \right)^{1/q},$$

where $\Gamma(P, Q)$ is the set of all couplings with marginals P and Q .

Advantage.

- It measures the minimum cost of transporting probability mass from one distribution to another.
- It captures geometric discrepancies in location, spread, shape, and support.

Method	Main idea	Limitation
Sliced Wasserstein	Average $W_1(v^\top X, v^\top Y)$ over many one-dimensional directions	Averaging may miss sparse signals
Projected Wasserstein	Optimize $W_q(A^\top X, A^\top Y)$ over low-dimensional subspaces A	Subspace search is harder; OT may still be costly
Smoothed Wasserstein	Compute $W_q(P * \mathcal{N}_\sigma, Q * \mathcal{N}_\sigma)$ after smoothing the two distributions	Smoothing may wash out local or sparse differences

Main Idea: Sparse Max-sliced Wasserstein

Key motivation

- Use low-dimensional projections to avoid high-dimensional transport.
- Avoid averaging away the few directions where the discrepancy is visible.
- Search for the projection where the two distributions differ the most.

The paper directly searches over sparse projection directions:

$$D(P, Q) = \sup_{v \in \mathcal{V}_{0,s}} W_1(v^\top X, v^\top Y),$$

$$\mathcal{V}_{0,s} = \{v \in \mathbb{R}^p : \|v\|_2 = 1, \|v\|_0 \leq s\}.$$

Interpretation. The informative direction may not be exactly s -sparse. However, searching over all high-dimensional directions is computationally and statistically difficult. The s -sparsity constraint is therefore a practical compromise: it reduces the effective search space, controls noise accumulation, and makes the direction optimization more tractable.

Dual View: Kantorovich–Rubinstein Formula

Kantorovich–Rubinstein formula

For a fixed direction v , let $Z = v^\top X$ and $U = v^\top Y$, with distributions P_v and Q_v . Then

$$W_1(P_v, Q_v) = \sup_{\|f\|_{\text{Lip}} \leq 1} \{\mathbb{E}f(Z) - \mathbb{E}f(U)\}.$$

Proof sketch. The primal transport problem is

$$W_1(P_v, Q_v) = \inf_{\gamma \in \Gamma(P_v, Q_v)} \int |z - u| d\gamma(z, u).$$

For any 1-Lipschitz f ,

$$\|f\|_{\text{Lip}} \leq 1 \implies f(z) - f(u) \leq |z - u|.$$

Thus, for any coupling γ ,

$$\mathbb{E}f(Z) - \mathbb{E}f(U) = \int \{f(z) - f(u)\} d\gamma(z, u) \leq \int |z - u| d\gamma(z, u).$$

Taking the infimum over γ gives weak duality. Since the Kantorovich problem is an **infinite-dimensional linear program**, Kantorovich duality gives strong duality and hence equality. $\square$

Why W_1 , not other Wasserstein distances?

Kantorovich–Rubinstein duality makes W_1 a supremum over 1-Lipschitz functions, so MSWD becomes an empirical-process statistic. For W_q with $q > 1$, the dual form is less tractable, so Gaussian approximation and bootstrap calibration do not apply directly.

Test Construction I: Standardized Statistic

The empirical sparse max-sliced Wasserstein distance is

$$D_n = \sup_{v \in \mathcal{V}_{0,s}} \sup_{f \in \text{Lip}_1} \left[\frac{1}{n_1} \sum_{i=1}^{n_1} f(v^\top X_i) - \frac{1}{n_2} \sum_{j=1}^{n_2} f(v^\top Y_j) \right].$$

For a fixed (f, v) , define

$$\theta_{f,v} = \mathbb{E} f(v^\top X) - \mathbb{E} f(v^\top Y).$$

The empirical difference fluctuates around $\theta_{f,v}$ at the two-sample rate:

$$\hat{\theta}_{f,v} = \theta_{f,v} + O_p \left\{ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)^{1/2} \right\}.$$

Therefore, we multiply D_n by the reciprocal rate

$$a_n = \left(\frac{n_1 n_2}{n_1 + n_2} \right)^{1/2}, \quad T_n = a_n D_n.$$

This scaling puts empirical noise on a stable $O_p(1)$ scale. Under H_0 , there is no population discrepancy, so T_n mainly measures sampling noise. Under H_1 , T_n also contains the scaled signal $a_n D(P, Q)$, and the test rejects when this signal exceeds the bootstrap-calibrated noise level.

Test Construction II: Bootstrap Critical Value

The null distribution of T_n is unavailable because the maximizing f and v are selected from the data. For each (f, v) , define the population and empirical discrepancies:

$$\theta_{f,v} = \mathbb{E}f(v^\top X) - \mathbb{E}f(v^\top Y), \quad \hat{\theta}_{f,v} = \frac{1}{n_1} \sum_{i=1}^{n_1} f(v^\top X_i) - \frac{1}{n_2} \sum_{j=1}^{n_2} f(v^\top Y_j).$$

The bootstrap uses independent multinomial weights $M \sim \text{Multinomial}(n_1; 1/n_1, \dots, 1/n_1)$ and $M' \sim \text{Multinomial}(n_2; 1/n_2, \dots, 1/n_2)$.

$$D^* = \sup_{v \in \mathcal{V}_{0,s}, f \in \text{Lip}_1} \left[\frac{1}{n_1} \sum_i (M_i - 1) f(v^\top X_i) - \frac{1}{n_2} \sum_j (M'_j - 1) f(v^\top Y_j) \right], \quad T^* = a_n D^*.$$

Basic-bootstrap insight

The centered weights $(M_i - 1)$ make the bootstrap mimic the first-order error

$$\hat{\theta}_{f,v} - \theta_{f,v} \quad \text{by} \quad \hat{\theta}_{f,v}^* - \hat{\theta}_{f,v}.$$

The goal is to approximate null fluctuation, not to reproduce the observed empirical distance itself.

Test Construction III: Simultaneous Confidence Intervals

Let α be the significance level, so $1 - \alpha$ is the confidence level. Let $\hat{q}_{T^*}(1 - \alpha)$ be the bootstrap $(1 - \alpha)$ -quantile of T^* . The bootstrap calibration targets

$$\Pr \left\{ \sup_{f,v} a_n(\hat{\theta}_{f,v} - \theta_{f,v}) \leq \hat{q}_{T^*}(1 - \alpha) \right\} \approx 1 - \alpha.$$

With probability about $1 - \alpha$, the following bound holds simultaneously for all (f, v) :

$$\theta_{f,v} \geq \hat{\theta}_{f,v} - \frac{\hat{q}_{T^*}(1 - \alpha)}{a_n}.$$

Since this is a simultaneous, intersection-type guarantee, each individual interval is also valid, usually conservatively. The one-sided SCI is

$$\text{SCI}(f, v) = \left[\hat{\theta}_{f,v} - \frac{\hat{q}_{T^*}(1 - \alpha)}{a_n}, \infty \right).$$

How to use it:

- For the maximizing pair $(\hat{f}, \hat{v})$, if $L_{\hat{f}, \hat{v}} > 0$, equivalently $0 \notin \text{SCI}(\hat{f}, \hat{v})$, reject H_0 .
- After rejection, use SCI for interpretation: $L_{f,v} > 0$ indicates a significant discrepancy along v . Here $v = e_k$ checks one coordinate, while sparse v 's check low-dimensional linear combinations.

Computation I: One-dimensional W_1 Closed Form

For univariate distributions F and G ,

$$W_1(F, G) = \int_{\mathbb{R}} |F(t) - G(t)| dt = \int_0^1 |F^{-1}(u) - G^{-1}(u)| du.$$

Key identities behind the formula:

$$|x - y| = \int_{\mathbb{R}} |\mathbf{1}\{x \leq t\} - \mathbf{1}\{y \leq t\}| dt, \quad \mathbb{E}|A - B| \geq |\mathbb{E}A - \mathbb{E}B|.$$

Lower-bound direction: With the identities above and Fubini, for any coupling (X, Y) ,

$$W_1(F, G) = \mathbb{E}|X - Y| = \int_{\mathbb{R}} \mathbb{E}|\mathbf{1}\{X \leq t\} - \mathbf{1}\{Y \leq t\}| dt \geq \int_{\mathbb{R}} |F(t) - G(t)| dt = \int_0^1 |F^{-1}(u) - G^{-1}(u)| du.$$

Upper-bound direction: With the quantile coupling

$$U \sim \text{Unif}(0, 1), \quad X = F^{-1}(U), \quad Y = G^{-1}(U),$$

we obtain

$$W_1(F, G) \leq \mathbb{E}|F^{-1}(U) - G^{-1}(U)| = \int_0^1 |F^{-1}(u) - G^{-1}(u)| du.$$

then the one-dimensional closed form follows.

Computational consequence

For fixed v , the best f does not need to be explicitly optimized. The one-dimensional W_1 can be computed by sorting or cumulative mass matching. Thus the main remaining problem is how to search over v .

Computation II: ℓ_1 Screening for ℓ_0 Sparsity

The original direction search is combinatorial:

$$\max_{\|v\|_2=1, \|v\|_0 \leq s} W_1(v^\top \mu_1, v^\top \mu_2).$$

The algorithm uses a sequential ℓ_1 approximation:

$$\mathcal{V}_{1,\lambda} = \{v : \|v\|_2 = 1, \|v\|_1 \leq \lambda\}.$$

- ① **ℓ_1 -screening:** scan $\lambda \in [1, \sqrt{s}]$ and use projected subgradient ascent on $\mathcal{V}_{1,\lambda}$. The support of v may change during this stage.
- ② **Filtering:** keep local maximizers $\hat{v}_\lambda$ satisfying $\|\hat{v}_\lambda\|_0 \leq s$, and choose the one with the largest empirical W_1 .
- ③ **Refitting:** set $S = \text{supp}(\hat{v})$, fix $v_{S^c} = 0$, and re-optimize the direction only within S .

Our interpretation

The ℓ_1 step is mainly a screening device: it promotes sparse candidate supports in a tractable continuous problem. After filtering candidates with $\|\hat{v}\|_0 \leq s$, the algorithm refits on the selected support S with $\lambda = \sqrt{|S|}$, which removes extra ℓ_1 shrinkage inside S .

Theoretical Validity I: Proof Ingredients

Goal. The theoretical difficulty is to control a supremum over the large class

$$\mathcal{G}_s = \{g(x) = f(v^\top x) : v \in \mathcal{V}_{0,s}, f \in \text{Lip}_1\}.$$

Assumptions.

- Sub-exponential tails: uniformly bounded ψ_1 -Orlicz norms.
- Balanced sample sizes: $n_1/(n_1 + n_2)$ bounded away from 0 and 1.
- Mild non-degeneracy for some projected distribution.

Main proof ingredients.

- 1 **Truncation.** Set $q_n = 2\nu \log(pn)$ and $\kappa_n = \sqrt{s} q_n = 2\sqrt{s}\nu \log(pn)$. With probability at least $1 - 2/n$, all sparse projected sample points lie in $[-\kappa_n, \kappa_n]$. Thus only the values of f on this bounded interval matter.
- 2 **ϵ -net discretization.** After truncation, approximate the infinite class by finite nets: choose sparse supports, net the unit sphere on each support, and discretize f on $[-\kappa_n, \kappa_n]$.
- 3 **Comparison tools.** After discretization, the proof applies:
 - **Gaussian approximation:** compare the empirical-process supremum with a Gaussian supremum.
 - **Anti-concentration:** convert distributional approximation into Kolmogorov-distance control.
 - **Bootstrap comparison:** show the conditional bootstrap supremum admits the same Gaussian approximation.

Theoretical Validity II: Gaussian Approximation Target

For $g \in \mathcal{G}_s$, define the centered two-sample empirical process

$$\mathbb{G}_n g = a_n \{ (P_n^X - P)g - (P_n^Y - Q)g \}, \quad a_n = \left(\frac{n_1 n_2}{n_1 + n_2} \right)^{1/2}.$$

Under $H_0 : P = Q$, $T_n = \sup_{g \in \mathcal{G}_s} \mathbb{G}_n g$.

Truncated class $\mathcal{G}_n$. Since $\mathcal{G}_s$ is not compact under $\|\cdot\|_\infty$, for $\kappa_n = 2s^{1/2} \nu \log(pn)$ define

$$\mathcal{F}_n = \{f \in \mathcal{F} : f(x) = f(-\kappa_n) \text{ if } x < -\kappa_n, f(x) = f(\kappa_n) \text{ if } x > \kappa_n\}, \quad \mathcal{G}_n = \{f \circ v : f \in \mathcal{F}_n, v \in \mathcal{V}_{0,s}\}.$$

By Lemma 2, with probability at least $1 - 2/n$: $\mathbb{G}_n \mathcal{G}_s = \mathbb{G}_n \mathcal{G}_n$.

Theorem 1 (Gaussian approximation). Let $Z = \{Z_g : g \in \mathcal{G}_n\}$ be the centered Gaussian process with covariance $\mathbb{E}\{Z(f)Z(g)\} = (n_1 + n_2)^{-1} [n_2 \text{cov}\{f(X), g(X)\} + n_1 \text{cov}\{f(Y), g(Y)\}]$. Then

$$d_K\{\mathcal{L}(\mathbb{G}_n \mathcal{G}_s), \mathcal{L}(Z\mathcal{G}_n)\} \rightarrow 0.$$

Theorem 2 (Bootstrap approximation). For $T^* = \sup_{g \in \mathcal{G}_s} G_n^* g$,

$$d_K\{\mathcal{L}(T^* | \mathcal{D}), \mathcal{L}(Z\mathcal{G}_n)\} \rightarrow 0 \quad \text{in probability.}$$

Implication. T_n and T^* are both linked to the *same* Gaussian supremum $Z\mathcal{G}_n$ (over the truncated class, not $\mathcal{G}_s$ directly), so the bootstrap quantile is an asymptotically valid critical value.

Theoretical Validity III: Asymptotic Size Control

Theorem 3 (Validity). Suppose Assumptions 1–3 hold. If $s \lesssim 1$ and $p \lesssim n^\gamma$ for any $0 \leq \gamma < \infty$ then, under H_0 ,

$$|\mathbb{P}\{T_n > q_{T^*}(1 - \alpha)\} - \alpha| \rightarrow 0,$$

where $q_{T^*}(\cdot)$ is the quantile function of T^* .

Why it holds.

- Direct corollary of Theorems 1 and 2: both $\mathcal{L}(\mathbb{G}_n \mathcal{G})$ and $\mathcal{L}(T^* | \mathcal{D})$ converge in Kolmogorov distance to the same Gaussian law $\mathcal{L}(Z\mathcal{G}_n)$.
- Hence $d_K\{\mathcal{L}(\mathbb{G}_n \mathcal{G}), \mathcal{L}(T^* | \mathcal{D})\} \rightarrow 0$ in probability: the bootstrap quantile is asymptotically exact for the true null quantile.

Implication.

- For any fixed sparsity level s and dimension growing polynomially in n , the Type-I error is asymptotically controlled at the nominal level α .
- Validity holds without restricting $p \leq n$; p may far exceed n .

Theoretical Validity IV: Power and Consistency

Theorem 4 (Consistency). Suppose Assumptions 1–2 hold and $n^{-1}s^2 \log^4(pn) \lesssim 1$. If, for some sufficiently large constant $c_a > 0$,

$$\sup_{g \in \mathcal{G}_s} [\mathbb{E}\{g(X)\} - \mathbb{E}\{g(Y)\}] \geq c_a n^{-1/2} \{s \log(pn)\}^{1/2},$$

then H_0 is rejected with probability tending to 1.

Reading the rate.

- The detectable signal may shrink with n at rate $n^{-1/2} \{s \log(pn)\}^{1/2}$, and the test remains consistent.
- Dependence on dimension p enters only through $\log(pn)$ — logarithmic, not polynomial.
- The condition $n^{-1}s^2 \log^4(pn) \lesssim 1$ holds automatically when $s \lesssim 1$ and $p \lesssim n^\gamma$.

Contrast with MMD (Remarks 2–3).

- MMD-based tests require the moment gap to grow *polynomially* in p , e.g. $\mu_P - \mu_Q \gtrsim n^{-1/2} p^{1/4}$ for nontrivial power.
- The proposed test avoids this p -vs- n tradeoff: power holds whenever the s -sparse max-sliced Wasserstein distance is adequately large.

Part II: Reproduction Study

Numerical simulation, bootstrap diagnostics, and methylation data analysis

Reproduction Overview

Goal. We reproduce the paper's experimental pipeline: first evaluate power under a controlled sparse signal, then inspect the bootstrap decision rule, and finally apply the method to real biomedical data analysis.

Experiment	Question	Output
Model A comparison	Does MSWD gain power under sparse decaying mean shifts?	Rejection curves from 50 runs
MSWD-only bootstrap	How does the observed statistic compare with its bootstrap null distribution?	T_n , threshold, p-value
LGG/GBM data	Do LGG and GBM differ in DNA methylation distributions, and where are the differences localized?	Reject H_0 ; 138 marginal related genes; 7 significant GO terms

Experiment 1 Setup: Model A

Data-generating model:

$$X \sim N(0_p, \Sigma), \quad Y \sim N(\mu, \Sigma),$$

where

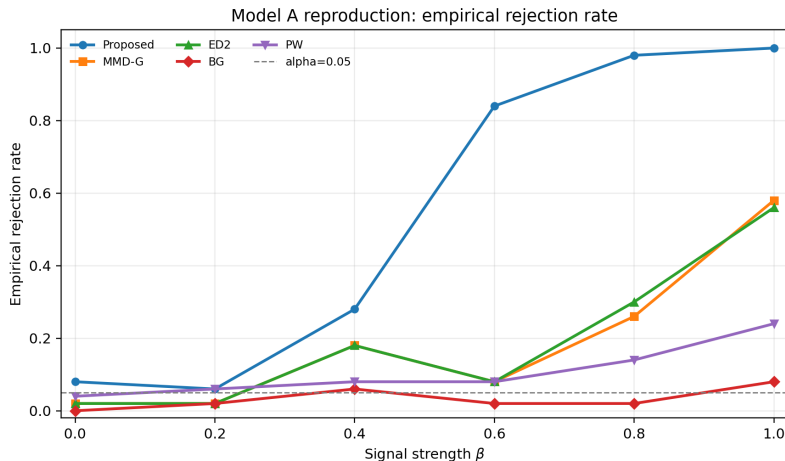
$$\Sigma_{ij} = 0.5^{|i-j|}, \quad \mu_j = 0.8\beta j^{-3}.$$

Parameters in reproduction:

- $n_1 = n_2 = 250$, $p = 500$, $\alpha = 0.05$.
- Signal grid: $\beta \in \{0, 0.2, 0.4, 0.6, 0.8, 1.0\}$.
- Proposed method uses bootstrap size $B = 300$.
- Compared methods: Proposed, MMD-G, ED2, BG, PW.
- MMD-L is not included because it was not recoverable from logs.
- Results are based on the first 50 complete runs for each β .

Why this model is relevant: although μ is formally nonzero in all coordinates, $\mu_j = 0.8\beta j^{-3}$ decays rapidly, so the distributional difference is concentrated in a few leading coordinates. This matches the setting where sparse projection search should be powerful.

Experiment 1 Result: Proposed Gains Power Fastest



The proposed MSWD test shows the fastest power growth as β increases. The advantage becomes clear from $\beta = 0.6$, where the sparse projected signal is strong enough to stand out from the bootstrap-calibrated noise.

Experiment 1 Interpretation

β	Proposed	MMD-G	ED2	BG	PW
0.0	0.08	0.02	0.02	0.00	0.04
0.2	0.06	0.02	0.02	0.02	0.06
0.4	0.28	0.18	0.18	0.06	0.08
0.6	0.84	0.08	0.08	0.02	0.08
0.8	0.98	0.26	0.30	0.02	0.14
1.0	1.00	0.58	0.56	0.08	0.24

- At $\beta = 0$, Proposed has rejection rate 0.08; with only 50 runs, Monte Carlo error is non-negligible.
- From $\beta = 0.6$, Proposed separates sharply from MMD-G, ED2, BG, and PW.
- This matches the paper's message: max-sliced search is effective for sparse or decaying high-dimensional signals.

Experiment 2: MSWD-only Bootstrap Diagnostic

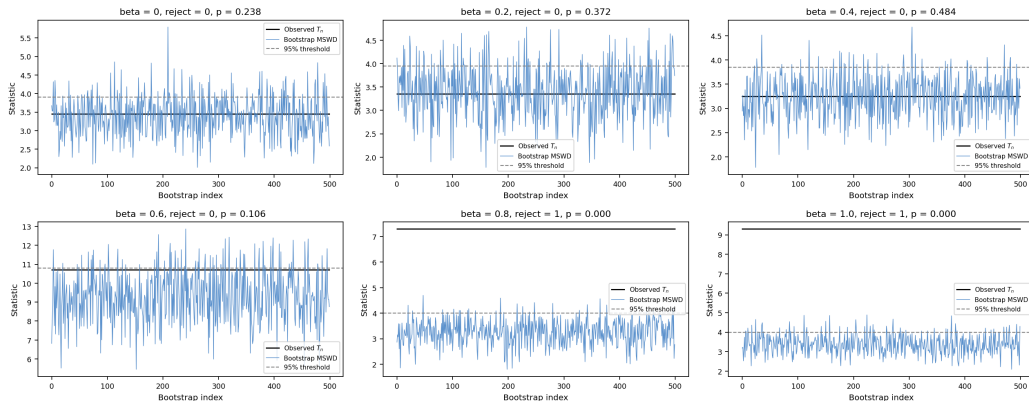
Setup. For each β , one Model A data set is generated and tested using $B = 500$ bootstrap replications. This is not a power experiment; it visualizes the test decision for one observed data set.

β	Observed T_n	Threshold	p-value	Reject
0.0	3.442	3.885	0.238	no
0.2	3.372	3.958	0.372	no
0.4	3.256	3.918	0.484	no
0.6	10.557	10.920	0.106	no
0.8	7.319	4.025	0.000	yes
1.0	9.479	3.921	0.000	yes

Interpretation. The bootstrap distribution estimates null fluctuation for the observed data. Rejection occurs when T_n exceeds the 95% bootstrap threshold. Here $\beta = 0.6$ is a borderline non-rejection in this single draw, while $\beta = 0.8$ and 1.0 clearly reject.

Experiment 2 Result: Strong Signals Exceed the Threshold

Model A: Observed MSWD Statistic vs Bootstrap Statistics



The bootstrap curve should not be interpreted as a power curve. It is a conditional simulation of the null distribution for the observed data set. When no bootstrap statistic exceeds the observed statistic, we report $p < 1/500 = 0.002$, rather than exactly zero.

Experiment 3 Background: LGG vs GBM DNA Methylation

Biomedical context. Glioma is a tumor arising from glial cells in the central nervous system. The case study compares two clinically different glioma groups:

- LGG: lower-grade glioma, usually less aggressive.
- GBM: glioblastoma, a highly aggressive glioma subtype.

Data. The paper analyzes TCGA DNA methylation data for glioma prognosis-related genes:

- LGG group: $n_1 = 511$.
- GBM group: $n_2 = 150$.
- Gene dimension: $p = 207$.
- Method: ℓ_0 -sparse MSWD with $B = 500$ bootstrap replications.
- Significance level: $\alpha = 0.05$.

Aim. Test whether LGG and GBM have different high-dimensional DNA methylation distributions, and then localize the difference to relevant marginal features and gene sets.

Experiment 3 Results: Global Rejection and Localization

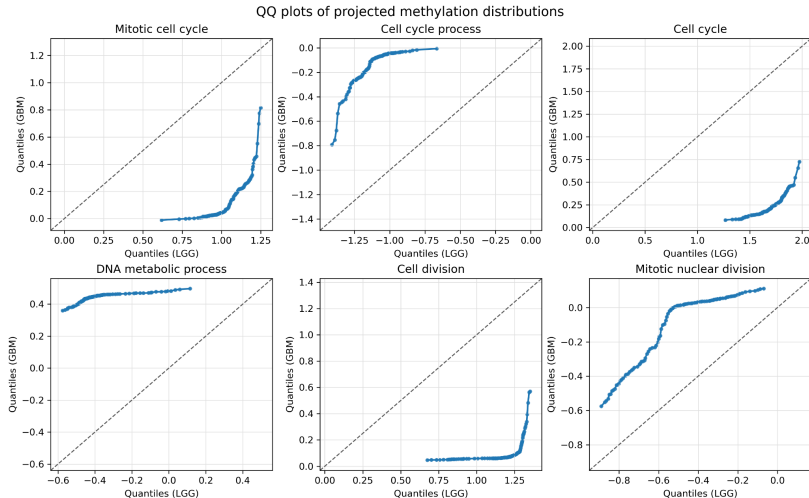
Global summary	
Metric	Value
Test statistic	42.9236
Bootstrap critical value	1.8479
Empirical p-value	$p < 0.002$
Sig. marginal genes	138/207
Sig. GO terms	7/7

GO-term localization		
GO term	Genes	Statistic
Mitotic cell cycle	5	9.467
Mitotic cell cycle process	4	9.465
Cell cycle process	7	10.196
Cell cycle	11	15.454
DNA metabolic process	6	8.730
Cell division	5	11.126
Mitotic nuclear division	3	4.131

Interpretation.

- The global statistic is far above the bootstrap critical value, so the test strongly rejects $H_0 : P = Q$.
- The reproduction finds 138 significant marginal genes, close to the 139 reported in the paper.
- All seven selected GO terms are significant, suggesting that the detected differences are mainly related to cell-cycle, DNA-metabolic, cell-division, and mitotic processes.

Experiment 3 QQ Plots: Projected Distributions



Each panel projects LGG and GBM samples along an optimized sparse direction for a GO term. Clear deviation from the identity line indicates distributional separation after projection. Orientation differences may occur because v and $-v$ represent the same projection direction up to sign.

Integrated Conclusion from Reproduction

- **Experiment 1:** MSWD has the strongest power under sparse decaying mean shifts.
- **Experiment 2:** bootstrap thresholds explain when the Proposed method rejects or fails to reject for a single observed data set.
- **Experiment 3:** LGG and GBM methylation distributions differ globally, marginally, and at the GO-term level.

Methodological message. Max-sliced search reduces signal dilution by looking for the most informative sparse projection, while the bootstrap and SCI framework unifies global testing with localization.

Limitations and Future Research (Stated by the Authors)

- **Data-dependent tuning.** The sparsity parameter s is selected via k -fold cross-validation, making s depend on the full data — formally at odds with the “ s fixed” premise underlying Theorems 1–4.
 - *Remedy considered:* sample splitting — use part of the data to tune s , the rest to test, then condition on the selected s .
 - *Not adopted in practice:* reduces effective sample size and finite-sample power; cross-validation already performs well numerically.
 - **Open problem:** rigorous theory for the data-driven tuning procedure is left for future work.
- **Two-sample scope.** The paper only treats $K = 2$ populations.
 - Extension to the K -sample (MANOVA-type) problem is suggested via the bootstrapping technique of Lin et al. (2023); unlike MMD, which does not extend intrinsically to $K > 2$.
 - A full theoretical treatment of the multi-sample case is left for future research.
- **Scope of Assumption 3.** Requires at least one projection direction with a density bounded away from zero on an interval; excludes the fully discrete case.
 - Relaxation to discrete coordinates is possible but “requires much more involved arguments without adding further insight,” so not pursued.

Further Critical Considerations

- **Computation–theory gap.** The sequential ℓ_1 approximation (screening $\rightarrow$ filtering $\rightarrow$ refitting) is a heuristic: it does not guarantee convergence to the global optimizer of the ℓ_0 -constrained direction search.
 - No theoretical result quantifies how this computational approximation error propagates into the power of the test.
- **Power under dense alternatives.** The ℓ_0 max-type statistic is expected to lose power against dense alternatives (differences spread across many coordinates).
 - Relative to averaging-based methods such as MMD or energy distance, no theoretical lower bound on this power degradation is given.

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Thank you for listening!